INCH-POUND

A-A-59372B
January 7, 2025
SUPERSEDING
A-A-59372A
June 25, 2004

COMMERCIAL ITEM DESCRIPTION

CALIBRATOR, OSCILLOSCOPE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This commercial item description (CID) describes an automated upgradeable oscilloscope calibration workstation with minimum amplitude of ± 200 Volts direct current (DC) into a 1 megaohm (MΩ) load with an accuracy of ± 0.025% + 25 microvolts (µvolts). This oscilloscope calibrator CID uses oscilloscope calibrator with active head for required specifications.

2. CLASSIFICATION. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following characteristics. The equipment shall be Class 4 in accordance with MIL-PRF-28800.

2.1. Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein. It shall be designed in a manner to be upgradeable as needed, to a minimum of 6.4 gigahertz (GHz) with enhanced parameters.

3.1. Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated in a temperature range of 5 to 40° Celsius (C).

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3.1.1 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of 0 to 50° C.

3.1.2 Humidity Range Operating. The equipment shall conform to its specified performance and accuracy while being operated in a non-condensing humidity range of < 90% relative humidity (RH) from 5 to 30° C and < 75% RH 30 to 40° C.

3.1.3 Humidity Range Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in a non-condensing humidity environment of =95% RH from 0 to 50° C.

3.1.4 Warm Up Time. This shall be = 20 minutes.

3.2 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 120 volts alternating current (AC) (±10%) or 240 volts AC (±10%) at line frequencies of 47.5 hertz (Hz) to 52.5 Hz and 57 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. The equipment shall also have power consumption equal to or less than 400 Volt-Amperes.

3.2.1 The AC characteristic of operation shall include leveled sinewave to a maximum of 1.1 GHz and rise time capability to < 150 picoseconds (ps).

3.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95%.

3.4 Pre-Conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

3.6 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm- up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

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3.6.1 DC Voltage.

3.6.1.1 Amplitude. This instrument shall provide the capability of producing a DC voltage for calibration purposes from ±1 millivolts direct current (mVDC) to ±200 VDC into a 1 MΩ load. Alternatively, this instrument shall provide the capability of producing a DC voltage for calibration purposes from ±1 mVDC to ±5 VDC into a 50 ohm (Ω) load.

3.6.1.2 Accuracy. This instrument shall provide a DC voltage accuracy of = ±0.025% + 25 µvolts.

3.6.1.3 Ranges. This instrument shall provide DC Volt/Div factors of 1, 2, 5 or 1, 2, 2.5, 4, 5, or continuously variable.

3.6.1.4 Deviation. This instrument shall provide a DC voltage deviation of = ±11.2%.

3.6.2 Squarewave Output Parameters.

3.6.2.1 Amplitude Parameters.

3.6.2.1.1 Range. This instrument shall provide Squarewave Output with an amplitude range of = 40 µV to = 200 V pk-pk into a 1 MΩ load and = 40 µV to = 5 V pk-pk into a 50Ω load.

3.6.2.1.2 Polarity. The instrument shall provide an Squarewave Output Polarity that may be positive, negative, or symmetrical about ground.

3.6.2.1.3 Ranging. The instrument shall provide Squarewave Output Ranging that shall be the same as stated in paragraph 2.7.1.3.

3.6.2.1.4 Deviation. The instrument shall provide Squarewave Output Deviation equal to that stated in paragraph 2.7.1.4.

3.6.2.2 Rise/Fall Time. This instrument shall provide a Squarewave Output Rise/Fall Time of < 150 nanoseconds (ns) for outputs < 100 V pk-pk, and < 200 ns for outputs of = 100 V pk-pk.

3.6.2.3 Aberrations. This instrument shall provide Squarewave Output Aberrations of < 2% peak for first 500 ns; < 0.1% peak after 500 ns; and < 0.01% peak after 100 µs.

3.6.2.4 Frequency Parameters.

3.6.2.4.1 Range. This instrument shall provide an Squarewave Output Frequency Range of 10 Hz to 100 kilohertz (kHz).

3.6.2.4.2 Accuracy. This instrument shall provide an Squarewave Output Frequency with an accuracy of ± 0.25 parts per million (ppm).

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3.6.2.4.3 Ranging. This instrument shall provide an Squarewave Output Frequency with ranging the same as that stated in paragraph 2.7.1.3.

3.6.3 Edge Functions. This instrument shall provide edge functions with the following parameters.

3.6.3.1 Low-Edge Pulse Parameters.

3.6.3.1.1 Rise/Fall Time 10% to 90%. This instrument shall provide a Rise/Fall Time on the Low-Edge Pulse of = 500 ps return to ground.

3.6.3.1.1.1 Rise/Fall Time. This instrument shall provide a Low-Edge Pulse Rise and Fall time of + 50 ps to – 150 ps.

3.6.3.1.2 Duty Cycle. This instrument shall provide a Low-Edge Pulse Duty Cycle of at least 10%.

3.6.3.1.3 Aberration into a voltage standing wave ratio (VSWR) of 1.2:1. This instrument shall provide Low-Edge Pulse aberrations, at the stated VSWR, that shall not exceed ±2% peak in 8 GHz or ±1.5% peak in 3 GHz for the first 10 ns.

3.6.3.1.4 Amplitude Parameters.

3.6.3.1.4.1 Range. This instrument shall provide a Low-Edge Pulse Amplitude Range that is = 5 millivolts (mV) pk-pk to = 3 V pk-pk into a 50Ω load.

3.6.3.1.4.2 Accuracy. This instrument shall provide a Low-Edge Pulse Amplitude Accuracy of ± 2%.

3.6.3.1.4.3 Ranging. This instrument shall provide Low-Edge Pulse Amplitude Ranging the same as that stated in paragraph 2.7.1.3.

3.6.3.1.5 Frequency Parameters.

3.6.3.1.5.1 Range. This instrument shall provide a Frequency Range of 10 Hz to 2 MHz on the Low-Edge Pulse signal.

3.6.3.1.5.2 Accuracy. This instrument shall provide a Frequency Accuracy of ± 0.25 ppm on the Low-Edge Pulse signal.

3.6.3.1.5.3 Ranging. This instrument shall provide a Low-Edge Pulse Frequency Ranging that shall be the same as that stated in paragraph 2.7.1.3.

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3.6.3.2 .High-Edge Pulse Parameters.

3.6.3.2.1 Rise/Fall Time 10% to 90%. This instrument shall provide a High-Edge Pulse Rise/Fall time of < 150 ns for an amplitude of < 100 V or < 200 ns for an amplitude of = 100 V.

3.6.3.2.2 Duty Cycle. This instrument shall provide a High- Edge Pulse Duty Cycle of 50%.

3.6.3.2.3 Aberration. This instrument shall provide a High- Edge Pulse Aberration of < ± 2% peak for the first 500 ns.

3.6.3.2.4 Amplitude Parameters.

3.6.3.2.4.1 Range. This instrument shall provide a High- Edge Pulse amplitude range of 1 mV pk-pk to 200 V pk-pk into a 1 MΩ load or 1 mV pk-pk to 5 V pk-pk into a 50Ω load.

3.6.3.2.4.2 Accuracy. This instrument shall provide a High- Edge Pulse Amplitude Accuracy of ± 3%.

3.6.3.2.4.3 Ranging. The instrument shall provide a High-Edge Pulse Amplitude ranging that shall be the same as that stated in paragraph 2.7.1.3.

3.6.3.2.5 Frequency Parameters.

3.6.3.2.5.1 Range. This instrument shall provide a High-Edge Pulse Frequency Range of on this type of signal 10 Hz to 100 kHz.

3.6.3.2.5.2 Accuracy. This instrument shall provide a High- Edge Pulse Frequency Accuracy of ± 0.25 ppm.

3.6.3.2.5.3 Ranging. This instrument shall provide a High-Edge Pulse Frequency Range that shall be the same as that stated in paragraph 2.7.1.3.

3.6.3.3 Fast-Edge Pulse Parameters.

3.6.3.3.1 Rise/Fall Time 10% to 90%. This instrument shall provide a Fast-Edge Pulse Rise/Fall time of < 150 ps return to ground.

3.6.3.3.1.1 Rise/Fall Time Accuracy. This instrument shall provide a Fast Edge Pulse Rise and Fall Time Accuracy of ± 25 ps.

3.6.3.3.2 Duty Cycle. This instrument shall provide a Fast-Edge Pulse Duty Cycle of 10%.

3.6.3.3.3 Aberration. This instrument shall provide a Fast-Edge Pulse Aberration < 3% peak in 8 GHz or < 2% peak in 3 GHz in the first instance.

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3.6.3.3.4 Amplitude Parameters.

3.6.3.3.4.1 Range. This instrument shall provide a Fast-Edge Pulse amplitude range of 5 mV pk-pk to 3 V pk-pk into a 50Ω load.

3.6.3.3.4.2 Accuracy. This instrument shall provide a Fast-Edge Pulse Amplitude Accuracy of ± 2%.

3.6.3.3.4.3 Ranging. The instrument shall provide a Fast-Edge Pulse Amplitude ranging that shall be the same as that stated in paragraph 2.7.1.3.

3.6.3.3.5 Frequency Parameters.

3.6.3.3.5.1 Range. This instrument shall provide a High-Edge Pulse Frequency Range of on this type of signal 10 Hz to 2 megahertz (MHz).

3.6.3.3.5.2 Accuracy. This instrument shall provide a High-Edge Pulse Frequency Accuracy of ± 0.25 ppm.

3.6.3.3.5.3 Ranging. This instrument shall provide a High-Edge Pulse Frequency Range that shall be the same as that stated in paragraph 2.7.1.3.

3.6.4 Timing Marker Function Parameters. This instrument shall provide functionality to supply timing markers with the following parameters.

3.6.4.1 Timing Markers Styles. The instrument shall provide the following types of markers: Square, Sinewave, Pulse, and Narrow Triangle. In addition, the provided markers shall have the following parameters.

3.6.4.2 Timing Marker Sinewave. The period on a sinewave marker shall be from 450 ps to 9.009 ns.

3.6.4.3 Timing Marker Square Wave.

3.6.4.3.1 Period. The Squarewave Timing Marker Period shall be = 10 ns to = 55 s.

3.6.4.3.2 Rise/Fall Time. The Squarewave Timing Marker Rise/Fall Time shall be = 2 ns.

3.6.4.4 Timing Marker Pulsed.

3.6.4.4.1 Period. The Period on a Pulsed Timing Marker shall be from 1.0 microsecond (µs) to 55 seconds (s).

3.6.4.4.2 Rise/Fall Time. The Pulsed Timing Marker Rise/Fall Time shall be = 2 ns.

3.6.4.5 Timing Markers Narrow Triangle.

3.6.4.5.1 Period. The Period on a Narrow Triangle Timing Marker shall be from 1.0 µs to 55 s.

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3.6.4.5.2 Rise/Fall Time. The Triangle Timing Marker Rise/Fall Time shall be = 3.5% of the period.

3.6.4.6 Timing Markers Ranging. The Timing Marker Time/div Ranging shall be 1, 2, 5 or 1, 2.5, 4, 5 or continuous for all of the Timing Marker Styles listed in paragraph 2.7.4.1 of this CID.

3.6.4.7 Timing Markers Timing Jitter. The Timing Marker Jitter for all Timing Marker Styles listed in paragraph 2.7.4.1 shall be < 10 ps pk-pk with time window < 100 ns, < 100 ps pk-pk with time window < 100 µs and < 1000 ps pk-pk with time window < 10 millisecond (ms).

3.6.4.8 Timing Markers Deviation. The Timing Marker Deviation for all Timing Marker Styles listed in paragraph 2.7.4.1 shall be ± 45% for the period.

3.6.4.9 Timing Markers Amplitude. The Timing Marker Amplitude for all Timing Marker Styles listed in paragraph 2.7.4.1 shall be 100 mV to 1 Volt pk-pk.

3.6.4.10 Timing Markers Sub-Division. This instrument shall provide functionality such that every tenth marker can be set to a higher amplitude for periods = 1 µs for all Timing Marker Styles listed in paragraph 2.7.4.1.

3.6.4.11 Timing Marker Timing Accuracy. This instrument shall provide a Timing Marker Timing Accuracy of = ± 0.25 ppm for all Timing Marker Styles listed in paragraph 2.7.4.1.

3.6.5 Leveled Sine Function Parameters.

3.6.5.1 Frequency Range. This shall be 0.1 Hz to 1.1 GHz.

3.6.5.2 Amplitude Peak-to-Peak. This shall be 5 mV to 5 Volts into a 50Ω Load From 0.1 Hz to 550 MHz. Additionally, this shall be 5 mV to 3 Volts from 550 MHz to 1.1 GHz.

3.6.5.3 Accuracy. This shall be ± 1.5% at any single reference rate from 50 kHz to 10 MHz.

3.6.5.4 Flatness Reference Frequency into VSWR of 1.6:1 (1.2:1). This shall be ± 2% from 0.1 Hz to 300 MHz, ± 3% from 300 MHz to 550 MHz, ± 4% from 550 MHz to 1.1 GHz.

3.6.5.5 Harmonic Purity. This shall be < -35 decibels relative to the carrier (dBc) on the second harmonic, < -40 dBc at the third harmonic and < -40 dBc for all other spurious signals.

3.6.6 Dual Sine Function Parameters.

3.6.6.1 Frequency Range. This shall be 0.1 Hz to 1.1 GHz Note.

3.6.6.2 Time Alignment. This shall be ± 25 ps any channel to any channel

3.6.7 Input Impedance Functions.

3.6.7.1 Resistance Measurement.

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3.6.7.1.1 Range. The Resistance Measurement Range shall be 10Ω to 150Ω, and 50 kiloohms (kΩ) to 12 megaohms (MΩ).

3.6.7.1.2 Accuracy. The resistance accuracy shall be ± .5 % from 10Ω to 40Ω, ± 0.1 % from 40Ω to 90Ω, ± 0.5 % from 90Ω to 150Ω, ± .5 % from 50 kΩ to 800 kΩ, ± 0.1 % from 800 kΩ to 1.2 MΩ, and ± 0.5 % from 1.2 MΩ to 12 MΩ.

3.6.7.2 Capacitance Measurement.

3.6.7.2.1 Range. The Capacitance Measurement Range shall be 1 power factor (pF) to 95 pF.

3.6.7.2.2 Accuracy. The Capacitance Measurement Accuracy shall be ± 2% ± 0.25 pF from 1 pF to 35 pF, and ± 3% ± 0.25 pF from 35 pF to 95 pF.

3.6.8 Pulse Width Function Parameters. This instrument shall provide Pulse Width Functionality with the following specifications.

3.6.8.1 Pulse Width. The Pulse Width shall be 1 ns to 100 ns.

3.6.8.2 Accuracy. The Pulse Accuracy shall be < 5% ± 200 ps.

3.6.8.3 Adjustment Resolution. The Pulse Width Adjustment Resolution shall be < 50 ps from 1 ns to 4 ns, < 250 ps from 4 ns to 20 ns, and < 1 ns from 20 ns to 200 ns.

3.6.8.4 Rise and Fall Times. The Pulse Width Rise/Fall Time shall be < 500 ps.

3.6.8.5 Aberrations. The Pulse Width Aberrations shall be less than ± 5% peak.

3.6.8.6 Stability. The Pulse Width Stability shall be < 10 ps pk-pk over 10 min/°C.

3.6.8.7 Jitter. The Pulse Jitter with respect to Trigger shall be < 5 ps pk-pk.

3.6.8.8 Frequency. The Pulse Frequency shall be 1 kHz to 1 MHz.

3.6.8.9 Amplitude. The Pulse Amplitude shall be 1 V pk-pk into 50Ω.

3.6.9 Other Output Parameters.

3.6.9.1 DC Current.

3.6.9.1.1 Amplitude. This shall be ± 100 micro amp (µA) to ± 100 milliamp (mA).

3.6.9.1.2 Accuracy. This shall be (0.25% + 0.5 µA).

3.6.9.2 Squarewave Current.

3.6.9.2.1 Amplitude. This shall be ± 100 µA to ± 100 mA pk-pk.

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3.6.9.2.2 Duty Cycle Symmetry. This shall be 50% centered about ground.

3.6.9.2.3 Rise Time and Aberrations. This shall be < 150 ns and < 2% peak.

3.6.9.3 Composite Video Parameters.

3.6.9.3.1 Amplitude. This shall be 1.0 V pk-pk.

3.6.9.3.2 Pattern (Full Raster). This shall be white, grey or black.

3.6.9.3.3 Sync Polarity. This shall be positive or negative.

3.6.9.3.4 Standards. This shall be 625 lines at 50 Hz or 525 lines at 60 Hz.

3.6.9.3.5 Trigger Output. This shall be composite sync or odd field start.

3.6.9.4 Overload Pulse.

3.6.9.4.1 Amplitude. The Overload Pulse Amplitude shall be 5 V to 20 V into 50Ω.

3.6.9.4.2 Polarity. The Overload Pulse Polarity shall be Positive or Negative.

3.6.9.4.3 Duration. The Overload Pulse Duration shall be 0.2 s to 100 s, subject to pulse energy limits.

3.6.9.4.4 Energy. The Overload Pulse Energy shall be 1.6 jules (J)to 50 J.

3.6.9.4.5 Power into 50 O. The Overload Pulse Power shall be 0.3 watts (W)to 8 W.

3.6.9.4.6 Trigger. This instrument shall have the functionality for a manual trigger of the Overload Pulse at a rate no less than 0.3 Hz.

3.6.9.5 Zero Skew.

3.6.9.5.1 Unadjusted Skew. This shall be < ± 50 ps channel to channel.

3.6.9.5.2 Adjusted Skew. This shall be < ± 5 ps channel to channel.

3.6.9.5.3 Skew Temperature Coefficient. This shall be < 0.2 ps/°C

3.6.9.5.4 Rise/Fall Time. This shall be = 500 ps.

3.6.9.5.5 Relative Jitter. This shall be < 7 ps pk-pk.

3.6.9.6 Input Leakage Function.

3.6.9.6.1 Open Circuit Output. This shall be Leakage < ± 50 picoamps.

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3.6.9.6.2 Short Circuit Output. This shall be Offset < ± 15 µV.

3.6.9.7 LF Linear Ramp.

3.6.9.7.1 Waveforms. This shall be 1 V pk-pk triangle symmetrical about ground.

3.6.9.7.2 Linearity. This shall be < ± 0.1 % deviation over 10 % to 90 %.

3.6.9.7.3 Ramp Times. These shall be 1 ms, 10 ms, 100 ms, or 1 s.

3.6.10 Dimensions.

3.6.10.1 Unit. The Main Unit dimensions shall not exceed 5.5 inches x 17 inches x 17.5 inches, H x W x D.

3.6.10.2 Head. The Active Head dimensions shall not exceed 2.6 inches x 1.3 inches x 5.6 inches, H x W x D. Note Active Head dimensions apply only if an Active head is required to achieve functionality.

3.6.11 Weight. The total equipment weight shall be no greater than 30 pounds.

3.6.12 Warranty Period. These shall be no less than one year.

3.7 Manuals. The equipment shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from https://quicksearch.dla.mil or the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 Federal Acquisition Regulation (FAR). Copies of the FAR may be obtained from https://www.acquisition.gov.

7.1.3 Key Words.

Amplitude
Frequency
Relative humidity
Squarewave
Voltage

MILITARY INTEREST

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

Agent:
Air Force - 110

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